Investigation 1 (Bill that's the wrong USB!)

In this investigation, my team has received reports that a malicious file was dropped onto a host by a malicious USB. They have pulled the logs suspected and have tasked you with running the investigation for it.

Question 1.

What is the full registry key of the USB device calling svchost.exe in Investigation 1?

Let's start by mapping the logs stored in .evtx file

```
PS C:\Users\user> Get-WinEvent -Path '.\Documents\TryHackMe!\Sysmon\investigation 1\Investigation-1.evtx'


   ProviderName: Microsoft-Windows-Sysmon

TimeCreated                     Id LevelDisplayName Message
-----------                     -- ---------------- -------
3/6/2018 12:57:51 AM             5
3/6/2018 12:57:51 AM             1
3/6/2018 12:57:51 AM             1
3/6/2018 12:57:51 AM             9
3/6/2018 12:57:51 AM             9
3/6/2018 12:57:51 AM             9
3/6/2018 12:57:51 AM            13
3/6/2018 12:57:51 AM            13
3/6/2018 12:57:50 AM             9
3/6/2018 12:57:50 AM             3
3/6/2018 12:57:48 AM             1
```

Knowing the Sysmon events IDs we could put attention to the ID number "13" this meaning that some registry value was modified ideal for this scenario, once we determined it the next step would be narrow down by clues and find our value as:

-FilterXPath "*[System/EventID=13 and EventData/Data[@Name='Image']='C:\Windows\System32\svchost.exe']"

- **-FilterXPath** → lets filter events directly in the XML structure (very powerful and efficient).
- The filter here means:
    - System/EventID=13 → only events with ID **13** (Sysmon: Registry value set).
    - EventData/Data[@Name='Image']='C:\Windows\System32\svchost.exe' → only keep events where the Image field equals svchost.exe.

So at this point, we are left with only the **registry modification events where svchost.exe is the process**.

```
PS C:\Users\user> Get-WinEvent -Path '.\Documents\TryHackMe!\Sysmon\investigation 1\Investigation-1.evtx' |
>> Where-Object { $_.Id -eq 13 -and $_.Message -match "svchost.exe" } |
>> Format-List TimeCreated, Id, Message
PS C:\Users\user>
PS C:\Users\user> Get-WinEvent -Path '.\Documents\TryHackMe!\Sysmon\investigation 1\Investigation-1.evtx' `
>> -FilterXPath "*[System/EventID=13 and EventData/Data[@Name='Image']='C:\Windows\System32\svchost.exe']" |
>> ForEach-Object {
>>     [xml]$xml = $_.ToXml()
>>     $xml.Event.EventData.Data | Where-Object { $_.Name -eq "TargetObject" } | Select-Object -ExpandProperty "#text"
>> }
HKLM\System\CurrentControlSet\Enum\WpdBusEnumRoot\UMB\2&37c186b&0&STORAGE#VOLUME#_??_USBSTOR#DISK&VEN_SANDISK&PROD_U3_CRUZER_MICRO&REV_8.01#4054910EF19005B3&0#\FriendlyName
PS C:\Users\user> |
```

And here is our registry key that was modified.

Moving to next one

What is the device name when being called by RawAccessRead in Investigation 1?

The key here is “RawAccessRead”, given that Sysmon ID 9 = RawAccessRead we can use it to filtered and get our answer

```
Get-WinEvent -Path '.\Documents\TryHackMe!\Sysmon\investigation 1\Investigation-1.evtx' `

-FilterXPath "*[System/EventID=9]" |

ForEach-Object {

  [xml]$xml = $_.ToXml()

  $xml.Event.EventData.Data | ForEach-Object {

    "$($_.Name) = $($_.'#text')"

  }

}
```

```
PS C:\Users\user>
PS C:\Users\user> Get-WinEvent -Path '.\Documents\TryHackMe!\Sysmon\investigation 1\Investigation-1.evtx' `
>> -FilterXPath "*[System/EventID=9]" |
>> ForEach-Object {
>>     [xml]$xml = $_.ToXml()
>>     $xml.Event.EventData.Data | ForEach-Object {
>>         "$($_.Name) = $($_.'#text')"
>>     }
>> }
UtcTime = 2018-03-06 06:57:51.070
ProcessGuid = {2ca4c7ef-396f-5a9e-0000-0010a06d0100}
ProcessId = 1388
Image = C:\Windows\explorer.exe
Device = \Device\HarddiskVolume3
UtcTime = 2018-03-06 06:57:51.054
ProcessGuid = {2ca4c7ef-396e-5a9e-0000-00104f270100}
ProcessId = 892
Image = C:\Windows\System32\svchost.exe
Device = \Device\HarddiskVolume3
UtcTime = 2018-03-06 06:57:51.023
ProcessGuid = {2ca4c7ef-396e-5a9e-0000-00104f270100}
ProcessId = 892
Image = C:\Windows\System32\svchost.exe
Device = \Device\HarddiskVolume3
UtcTime = 2018-03-06 06:57:50.992
ProcessGuid = {2ca4c7ef-396e-5a9e-0000-00104f270100}
ProcessId = 892
Image = C:\Windows\System32\svchost.exe
Device = \Device\HarddiskVolume3
PS C:\Users\user>
```

With this script we can obtain the name of the device when was called by “RawAccessRead”

Let’s move to next question.

What is the first exe the process executes in Investigation 1?

The key here is process creation that is ID 1, so use it to narrow down our result

PS C:\Users\user> Get-WinEvent -Path '.\Documents\TryHackMe!\Sysmon\investigation 1\Investigation-1.evtx' `

>> -FilterXPath "[System/EventID=1]" |*

>> ForEach-Object {

>>   [xml]$xml = $_.ToXml()

>>   [PSCustomObject]@{

>>   TimeCreated = $_.TimeCreated

>>   Process   = ($xml.Event.EventData.Data | Where-Object { $_.Name -eq "Image" }).'#text'

>>   }

>> } |

>> Sort-Object TimeCreated |

>> Select-Object -First 1

I combined with time created and sort by first one

```
PS C:\Users\user> Get-WinEvent -Path '.\Documents\TryHackMe!\Sysmon\investigation 1\Investigation-1.evtx' `
>> -FilterXPath "*[System/EventID=1]" |
>> ForEach-Object {
>>     [xml]$xml = $_.ToXml()
>>     [PSCustomObject]@{
>>         TimeCreated = $_.TimeCreated
>>         Process     = ($xml.Event.EventData.Data | Where-Object { $_.Name -eq "Image" }).'#text'
>>     }
>> } |
>> Sort-Object TimeCreated |
>> Select-Object -First 1

TimeCreated          Process
-----------          -------
3/6/2018 12:57:48 AM C:\Windows\System32\WUDFHost.exe

PS C:\Users\user>
```

The script output shows this result "WUFHost.exe" but this actually is a normal services running in a machine, so we let's try with more result to find for something more suspicious

```
PS C:\Users\user> Get-WinEvent -Path '.\Documents\TryHackMe!\Sysmon\investigation 1\Investigation-1.evtx' `
>> -FilterXPath "*[System/EventID=1]" |
>> ForEach-Object {
>>     [xml]$xml = $_.ToXml()
>>     [PSCustomObject]@{
>>         TimeCreated = $_.TimeCreated
>>         Process     = ($xml.Event.EventData.Data | Where-Object { $_.Name -eq "Image" }).'#text'
>>     }
>> } |
>> Sort-Object TimeCreated |
>> Select-Object -First 4

TimeCreated          Process
-----------          -------
3/6/2018 12:57:48 AM C:\Windows\System32\WUDFHost.exe
3/6/2018 12:57:51 AM C:\Windows\System32\rundll32.exe
3/6/2018 12:57:51 AM C:\Windows\System32\calc.exe


PS C:\Users\user>
```

This time the output shows 2 results more, and we should pay attention to rundll32.exe given that is a utility often used to run dll in windows machines, that is very suspicious in this context.